

MEASUREMENT-GATED PROVENANCE ATTENUATION FOR FROZEN EEG REPRESENTATIONS

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ABSTRACT

Frozen EEG representations can encode both neural activity and signatures of the acquisition pipeline. Source predictability alone does not identify what should be removed: it can reflect measurement effects or genuine biological and population differences. We study how measurement calibration can guide reusable correction of a pretrained encoder’s representations on downstream datasets, without retraining the encoder. We introduce Measurement-Gated Provenance Attenuation (MGPA), a framework for selective, measurement-informed correction. Paired measurement contrasts define an editable subspace without requiring its wholesale removal. Correction targets account for source information in the coordinates left unchanged. For exact affine score matching, we show that different targets retain the same information at different movement costs, and that the conditional mean source score given the preserved coordinates minimizes expected cost within the analyzed class. We realize this principle in two complementary forms: Closed-form MGPA estimates a single analytic correction from measurement pairs, while Iterative MGPA uses learned nonlinear source critics. Both apply to individual representations without source identity at deployment. Controlled comparisons show stronger source attenuation with measurement-derived edit spaces than with rank-matched PCA and random subspaces at matched movement budgets. On real EEG recordings, Iterative MGPA reduces source accessibility while maintaining or improving observed task performance; a critic ablation demonstrates the additional value of nonlinear correction. Adapters fitted and selected on P300 improve the robustness of frozen N170 and MMN heads under device–label association shifts without target-task adapter tuning. Separate control-head performance remains close to identity.

1 INTRODUCTION

Frozen EEG foundation models enable lightweight transfer when labeled data are scarce (Wang et al., 2024), but freezing also preserves acquisition sensitivities already encoded in their representations. We ask how measurement calibration can guide reusable correction of these representations on downstream data, without retraining the encoder. Physiological waveforms are measurements, not direct observations of physiology. Reference, montage, channel configuration, preprocessing, and recording workflow can all alter the signal presented to the encoder (Nunez & Srinivasan, 2006; Dong et al., 2019). Acquisition signatures can help reconstruct a recording without being the physiological evidence a downstream task seeks.

When recording source (such as the acquisition device or preprocessing condition) correlates with a downstream label, a lightweight head may exploit these cues rather than task-relevant neural information. Strong average performance can conceal this shortcut until the source–target association changes (Geirhos et al., 2020), as related failures in multi-site clinical prediction illustrate (Zech et al., 2018). Such failure need not require an unseen device: changing how familiar acquisition conditions co-occur with the target can invalidate the same predictor. A source probe establishes accessibility, not whether a task head uses that channel. Moreover, source can be associated with genuine biological or population differences. Indiscriminate source erasure can therefore discard the very variation a downstream task needs (Yamashita et al., 2019).

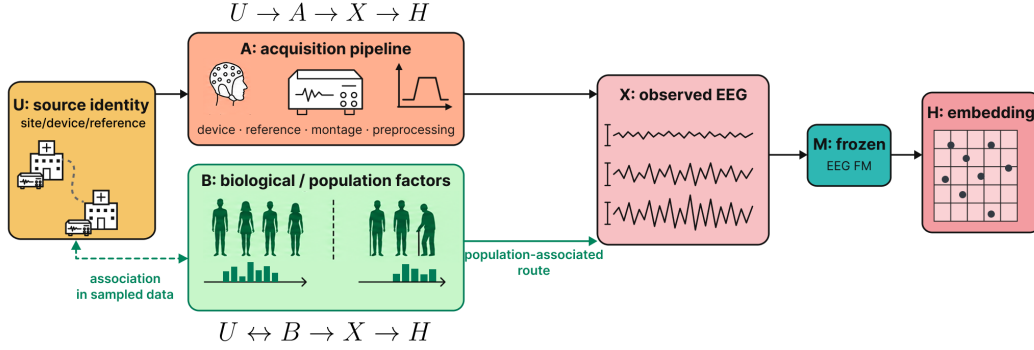


Figure 1: **Two routes to source-predictive frozen EEG representations.** Source predictability can arise through measurement A or sampling association with biological/population factors B . Here U denotes source, X observed EEG, and $H = M(X)$ its representation from frozen encoder M . Symbols are local to this diagram.

Source-labeled representations alone cannot generally separate the two routes in Figure 1. Measurement contrasts supply different evidence: they compare representations of the same underlying observation under specified measurement changes. Examples include applying different references to a stored EEG trial and observing a shared event with simultaneous recording systems. Such calibration provides evidence about which representation directions respond to measurement changes, rather than merely distinguish source groups.

We introduce *Measurement-Gated Provenance Attenuation* (MGPA), a principle and construction for selective correction of fixed representations. Its *measurement gate* restricts edits to a fixed subspace defined by measurement contrasts. The remaining coordinates Q stay unchanged. The gate permits changing a direction but does not require deleting it. Measurement-sensitive coordinates may still contain variation useful to downstream tasks. MGPA separates where measurement evidence permits a change from how each observation should move within that space.

We instantiate this principle in two complementary forms. *Closed-form MGPA* estimates a correction direction from paired measurement contrasts and applies one analytic adjustment to each representation. *Iterative MGPA* fits predictors of recording source, called *critics*, and uses their scores to guide small, bounded steps within the permitted subspace. Critics are refitted as representations change: *the critic proposes, and the intervention constrains*. Both implementations leave Q unchanged.

This leaves a shared question: what should correction target when Q remains unchanged? Suppose a predictor using Q alone assigns probability .8 to one of two equally likely sources. Editing the other coordinates cannot remove that retained evidence. Forcing a predictor using the full representation to output .5 instead asks editable coordinates to counteract evidence still available in Q .

Our analysis separates retained information from correction cost. For a fixed source score affine in the editable coordinates and a fixed correction metric, exact matching to different targets determined by Q preserves the same information at different movement costs. Expected cost is minimized by the conditional mean: the expected source score given Q , not the probability of source given Q . Both implementations estimate this conditional-mean target.

At fixed hyperparameters, both implementations are fitted using measurement calibration and source labels, without downstream task labels. In our reuse experiment, correction strength is selected using P300 development labels and held fixed across the other tasks. At deployment, the fitted map corrects individual representations without changing their dimension or requiring source identity, paired observations, or encoder retraining. The construction applies to fixed representations, including pretrained EEG embeddings and conventional EEG features.

Contributions. First, we formulate MGPA, combining measurement-supported edit permissions with selective correction that preserves a fixed complement. Second, we establish cost-optimal conditional-mean anchoring within the analyzed class of exact affine score-matching corrections and characterize Gaussian selective erasure, including residual source separation under direction error.

Third, we develop Closed-form MGPA and Iterative MGPA, two complementary implementations with source-blind deployment. Finally, we test selective correction, nonlinear critics and cross-task reuse in complementary controlled and recorded-data protocols. Table 1 previews the empirical evidence before the formal construction; Section 6 develops the full comparisons.

Table 1: **Main experimental questions and findings.** Source AUROC: lower is better; task AUROC: higher is better. Arrows compare identity with MGPA; row 3 reports gains over LEACE. Full comparisons and paired intervals are in Sections 6.2–6.4.

Research question and construction	Empirical preview
Can measurement-related source information be attenuated without losing task utility? Controlled SSVEP; Closed-form MGPA	Source AUROC: .683 $\rightarrow$.510. Task AUROC, head unchanged: .7528 $\rightarrow$.7526.
Does selective correction remain useful on real recordings? Temporal N170 and recorded wet/dry SSVEP; Iterative MGPA	N170: source AUROC .989 $\rightarrow$.946; refitted-task AUROC .728 $\rightarrow$.746. SSVEP: source AUROC .916 $\rightarrow$.872; refitted-task AUROC .688 $\rightarrow$.689.
Can a correction selected on one task improve shortcut robustness on other tasks? P300-selected Iterative MGPA	Frozen-head AUROC gains over LEACE at the worst device-label association: +.057 on N170 and +.019 on MMN.

2 RELATED WORK

Concept erasure and conditional invariance. INLP removes linearly predictive directions through repeated nullspace projection (Ravfogel et al., 2020); LEACE gives a closed-form, minimum-distortion affine eraser with linear guardedness (Belrose et al., 2023). Closed-form MGPA also yields an affine map, but estimates its correction direction from paired measurement contrasts and conditions its target on the coordinates preserved by the gate. IGBP extends erasure to nonlinear decision surfaces through iterative critic refitting (Iskander et al., 2023), while MANCE restricts nonlinear edits to local tangents of the natural representation manifold (Avitan et al., 2026). MGPA separates measurement-supported edit permissions from the correction rule: a source-predictive direction alone does not authorize its removal. This also differs from preserving covariance with a supplied task label, as in SPLINCE (Holstege et al., 2025). CIRCE learns nuisance independence conditional on task information (Pogodin et al., 2023), and conditional probing measures information beyond a baseline (Hewitt et al., 2021). In MGPA, the conditioning variable is the complement fixed by the measurement gate. Our analysis separates retained information from coordinate distortion, and its Gaussian specialization connects conditional-mean anchoring to minimum-distortion linear erasure.

Measurement harmonization and EEG transfer. Traveling-subject designs distinguish measurement bias from population sampling bias (Yamashita et al., 2019). Noisy Counterfactual Matching estimates a subspace from paired differences and removes it by projection (Bai et al., 2025). MGPA instead uses measurement contrasts to constrain selective edits. Several alignment methods need no target-task labels: CORAL matches domain covariances (Sun et al., 2016), Euclidean alignment normalizes EEG covariance (He & Wu, 2020), and CMMN aligns spectral distributions (Gnassounou et al., 2023). Riemannian Procrustes combines geometric alignment with a class-informed rotation (Rodrigues et al., 2019). AWar and SDDA address cross-headset classification through adaptation regularization and spatial distillation/distribution alignment, respectively (Wu et al., 2016; Liu et al., 2026). Both MGPA implementations operate on fixed representations, require no downstream labels during fitting at fixed settings, and apply a stored correction to individual representations without source identity.

Paired and post-training embedding adapters. FEATMAP is a close acquisition-harmonization predecessor: paired measurements train a reusable affine map between medical foundation-model embedding domains without downstream labels (Donle et al., 2026). Post-Training Augmentation Invariance learns lightweight adapters on frozen representations while approximately preserving the

geometry of unaugmented features (Eikenberry et al., 2026); PISCO separates style and content in pretrained representations (Ngweta et al., 2023). These works establish reusable post-training correction; MGPA specifies which changes calibration supports and how to perform them while retaining the fixed complement. MGPA links measurement-supported edit constraints to target choice. Closed-form MGPA implements this construction analytically; Iterative MGPA uses refitted source critics and bounded trust-region updates. Iterative MGPA’s local trust-region solve uses classical optimization machinery (Moré & Sorensen, 1983). Our comparisons distinguish edit permissions, calibration inputs and deployment routing, as well as source attenuation and downstream utility.

3 THE MGPA PRINCIPLE: MEASUREMENT-SUPPORTED CORRECTION

Fixed representations and measurement contrasts. Let $H \in \mathbb{R}^d$ be a representation produced by a frozen encoder or a fixed feature extractor. Its source $U \in \{0, 1\}$ indexes an acquisition system or a constructed measurement view; Y denotes a downstream task. Calibration data include paired representations $H_i^{(a)}, H_i^{(b)}$: either two specified transformations of the same stored trial or observations of the same unit under different acquisition conditions. A same-unit contrast supplies evidence about measurement response beyond observational source prediction. Assume paired views share a content contribution and have additive measurement responses in a fixed subspace. Their contrast $\Delta_i^{ab} = H_i^{(a)} - H_i^{(b)}$ cancels the shared contribution; the remaining response may depend on content in both magnitude and direction. Recovering the full response span requires calibration contrasts that span it (Bai et al., 2025).

A fixed complement, not wholesale deletion. Let $B \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the directions supported by calibration. The *measurement gate* $P = BB^\top$ restricts edits to their span. This constraint does not separate exclusively biological from instrumental coordinates. Let $Z = B^\top H$ be editable coordinates and $Q = q(H) = (I - P)H$ the fixed complement. Admissible corrections have the form

$$R(h) = q(h) + B\psi(h), \quad (1)$$

where $\psi(h) \in \mathbb{R}^r$ contains the corrected editable coordinates. Thus Q is recoverable from every admissible output. Conditioning on the fitted gate, the information chain rule gives

$$I\{U; R(H)\} = I(U; Q) + I\{U; R(H) \mid Q\}. \quad (2)$$

The first term is fixed by the admissible class. It can be nonzero through source-associated content or measurement variation outside the fitted gate; balanced source counts alone do not make the complement source-independent. The editable coordinates can also contain useful task variation, so permission to change them is not an instruction to discard them. This defines the correction problem: change selected components of Z while retaining Q .

A measurement-gate estimator. We estimate the measurement gate using normalized-contrast SVD. Fit-only preprocessing establishes shared featurewise-standardized coordinates. Each retained nonzero contrast Δ_i^{ab} is normalized to unit length. This makes the estimated subspace reflect recurring directions of measurement change rather than the magnitude of individual displacements. Stack these normalized contrasts as rows of D_A and compute its thin singular-value decomposition:

$$D_A = L_A \Sigma_A V_A^\top, \quad B = [v_1, \dots, v_r], \quad P = BB^\top. \quad (3)$$

The retained rank r is set by a retained-energy criterion or selected on development data, as specified by the experimental protocol. The gate remains fixed throughout fitting and deployment of the correction; a zero-rank gate returns identity. Appendix B specifies contrast construction, preprocessing, numerical filtering and rank selection.

From edit permission to provenance correction. The gate specifies where correction may act, but leaves its target and update rule to be chosen. We next study how targets determined by Q affect retained information and movement for a fixed affine source score. This analysis motivates the score’s conditional expectation given Q as a reference; for affine scores, that expectation can be obtained from $\mathbb{E}[Z \mid Q]$.

4 WHY THE GATE CHANGES THE TARGET

Figure 2 previews three choices: where edits are permitted, how much of the editable block to replace, and where to place the result. The gate fixes the first; we now study the other two.

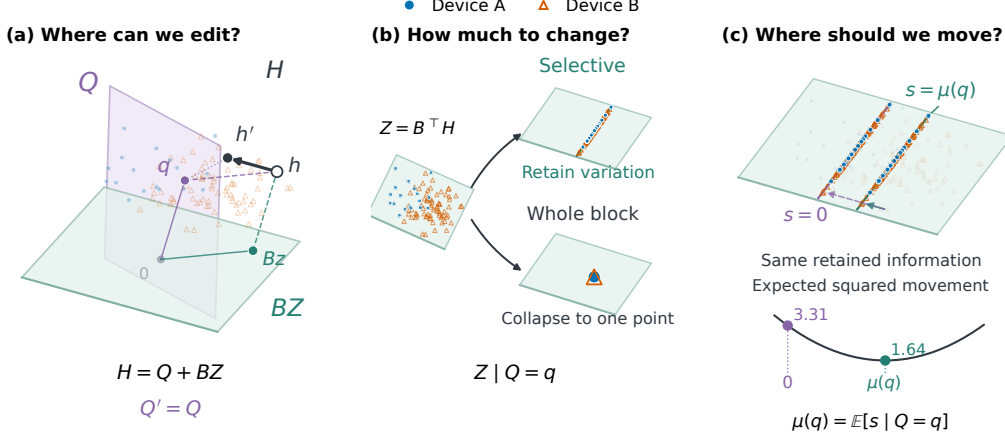


Figure 2: **Three choices in measurement-gated correction.** (a) Full embeddings $H = Q + BZ$: dashed guides show component projections; the black arrow edits BZ while preserving Q . The subspaces are orthogonal before the display projection. (b) At fixed $Q = q$, selective correction retains variation lost by replacing the whole editable block. (c) In the same conditional slice, two score targets retain the same information at different expected movement costs; the minimum is at the conditional-mean score $\mu(q)$. In this Gaussian illustration, Z has source means $(\pm 1, 0)$, covariance I_2 , and $\Pr(B | q) = .8$, giving $s = \log 4 + 2z_1$. Both targets make the source-conditional output laws coincide. Blue/orange denote devices; dots and costs are illustrative, not EEG results.

What should correction target when source evidence in Q must remain unchanged? The standard Bayes-odds decomposition separates this preserved evidence from the contribution of the editable coordinates. For $0 < p(q) = \Pr(U = 1 | Q = q) < 1$, write $s_Q(q) = \log p(q)$. When the conditional source laws are mutually absolutely continuous, the finite Bayes source score under the original representation distribution satisfies

$$s(q, z) = s_Q(q) + \log \frac{dP_{Z|Q=q,U=1}}{dP_{Z|Q=q,U=0}}(z). \quad (4)$$

Driving this fixed score to zero asks editable coordinates to offset the evidence carried by Q , even though that evidence remains available in the corrected representation. We therefore compare targets determined by Q , separating their effects on retained information from their movement costs.

4.1 FIXED-SCORE GEOMETRY: SAME INFORMATION, DIFFERENT MOVEMENT

To study target choice without requiring a Bayes-optimal predictor, fix a score $s(q, z) = b(q) + w(q)^\top z$, affine in the editable coordinates, and a positive-definite correction metric $\Omega(q)$. We condition on the fitted gate and use nonredundant coordinates for Q . The functions b, w, Ω may depend on q , but not on z , and $w(q) \neq 0$. For a finite measurable target $a(q)$, define

$$d_\Omega = w^\top \Omega^{-1} w, \quad v = \Omega^{-1} w / d_\Omega, \quad T_a(q, z) = (q, z + v\{a(q) - s(q, z)\}). \quad (5)$$

Standard metric projection gives the minimum-cost edit reaching this target. The following proposition combines this geometry with the conditional squared-error decomposition to characterize what target choice changes.

Proposition 1 (Target choice changes distortion, not retained information). *T_a is the unique minimum- Ω -norm edit reaching score $a(q)$. For any joint law, all such targets generate the same information: $\sigma(T_a(Q, Z)) = \sigma(Q, N)$, where $N = (I - vw^\top)Z$. Let $C_a(q)$ be the expected squared*

$\Omega(q)$ -norm of the editable displacement conditional on $Q = q$. With finite conditional score second moments,

$$C_a(q) = \frac{\text{Var}(s \mid Q = q) + \{\mathbb{E}[s \mid Q = q] - a(q)\}^2}{d_\Omega(q)}. \quad (6)$$

Consequently, $\mu(q) = \mathbb{E}[s \mid Q = q]$ uniquely minimizes expected squared movement in the chosen metric within this fixed-score family.

The editable output is $N + v(a - b)$, a translation of N determined by the retained Q . Every target in this fixed-score family therefore preserves the same information about any downstream variable, although a fixed task head may respond differently to the resulting coordinates.

The argument requires neither Gaussian data nor a Bayes-optimal score. With Euclidean movement, $\Omega = I$ and $d_\Omega = \|w\|^2$, giving the same conditional-mean optimum for exact affine score matching. Conditional source erasure holds precisely when $N \perp U \mid Q$ (Appendix A.2). The next subsection examines this condition in a Gaussian model.

4.2 GAUSSIAN CASE: WHEN SELECTIVE CORRECTION ACHIEVES CONDITIONAL ERASURE

When does this projection remove source information beyond Q , and what extra movement does zero-score targeting require? An equal-covariance Gaussian model makes both questions explicit. For $0 < p(q) < 1$, consider

$$Z \mid Q = q, U = u \sim \mathcal{N}(m(q) + (2u - 1)\delta/2, S), \quad S \succ 0, \quad D = \delta^\top S^{-1} \delta > 0. \quad (7)$$

Here $m(q)$ is the source-mean midpoint, whereas $\mathbb{E}[Z \mid Q = q] = m(q) + (2p(q) - 1)\delta/2$. The Bayes score is $s = s_Q + \ell$, with $\ell = \delta^\top S^{-1}(z - m(q))$. For a finite measurable target $a(q)$, write $\kappa(q) = a(q) - s_Q(q)$. Applying Proposition 1 with $w = S^{-1}\delta$ and $\Omega = S^{-1}$ gives $v = \delta/D$ and

$$R_\kappa(q, z) = \left(q, z + \frac{\delta}{D} \{ \kappa(q) - \ell(q, z) \} \right). \quad (8)$$

Zero-score targeting sets the original total score to zero and therefore uses $\kappa(q) = -s_Q(q)$. At each fixed q , the minimum-distortion affine eraser in this model specializes the binary LEACE geometry (Belrose et al., 2023). We use this specialization to compare the movement costs of different score targets.

Theorem 2 (Gaussian conditional erasure and target-dependent cost). *Under (7), every R_κ preserves Q and satisfies $R_\kappa(Q, Z) \perp U \mid Q$. Its conditional expected squared movement in the S^{-1} metric is*

$$C_\kappa(q) = 1 + Dp(1 - p) + \frac{\{\kappa(q) - (2p - 1)D/2\}^2}{D}, \quad p = p(q). \quad (9)$$

The minimizing residual target and corresponding total-score target are

$$\begin{aligned} \kappa_*(q) &= (2p - 1)D/2 = \mathbb{E}[\ell \mid Q = q], \\ a_*(q) &= s_Q(q) + \kappa_*(q) = \mathbb{E}[s \mid Q = q]. \end{aligned}$$

Relative to this target, zero-score targeting incurs the excess cost

$$C_{-s_Q}(q) - C_{\kappa_*}(q) = \frac{\{s_Q(q) + (2p - 1)D/2\}^2}{D} \geq 0. \quad (10)$$

At $p(q) = 1/2$, the two targets coincide. For fixed $p(q) \neq 1/2$, the excess diverges as $D \downarrow 0$. Among Q -preserving deterministic maps whose editable output is affine in Z at each fixed q , mean anchoring minimizes expected squared movement in the S^{-1} metric subject to conditional source erasure.

Conditional erasure does not require replacing the whole editable block by $\mathbb{E}[Z \mid Q]$. Full replacement costs $r + Dp(1 - p)$, versus $1 + Dp(1 - p)$ for selective mean anchoring, and discards $r - 1$ additional conditionally source-independent directions that may carry task information. This illustrates why edit permission need not imply wholesale deletion; Appendix A.4 gives the proof and cost comparison.

The preceding result uses the correct direction. What remains when calibration supplies only an estimate?

Corollary 3 (Residual separation from direction error). *Condition on a fitted gate and correction obtained independently of the evaluation observation, and assume (7) holds in these coordinates. In (5), take $w = C^{-1}\hat{\delta}$, $\Omega = C^{-1}$, with $C \succ 0$, $\hat{\delta} \neq 0$, and any finite Q -dependent target. Let D_{out} denote the squared Mahalanobis separation of the two corrected source laws conditional on $Q = q$, using their common output covariance on its support. Then*

$$\frac{D_{\text{out}}}{D} = 1 - \frac{(\delta^\top S^{-1} \hat{\delta})^2}{D \hat{\delta}^\top S^{-1} \hat{\delta}} = \sin^2 \angle(S^{-1/2} \delta, S^{-1/2} \hat{\delta}). \quad (11)$$

These results separate three design choices. The gate fixes the coordinates that correction must preserve. For a full analytic step in the Gaussian model, direction error determines residual source separation, while the fitted metric and target affect movement and output coordinates. Conditional-mean anchoring minimizes expected movement within the fixed-score family; estimating this anchor introduces an additional cost. Appendices A.5–A.6 provide the direction-error proof and anchor-error identity.

5 CONSTRUCTING THE MGPA CORRECTION

Closed-form MGPA implements the affine projection with an estimated direction, metric, and conditional-mean anchor. Iterative MGPA carries the gate and conditional-target principles into bounded local updates driven by learned source scores.

An analytic correction from measurement pairs. Closed-form MGPA estimates its direction $\hat{\delta}$ as the mean calibration contrast $B^\top (H_i^{(b)} - H_i^{(a)})$. On the fitting data, we estimate the pooled conditional mean $\hat{m}_Z(q) \approx \mathbb{E}[Z \mid Q = q]$ by ridge regression and a regularized covariance $C \succ 0$ from residuals of a joint regression of Z on Q and source label U . For a nonzero fitted direction, Closed-form MGPA applies

$$z' = z - \alpha \hat{\delta} \frac{\hat{\delta}^\top C^{-1} [z - \hat{m}_Z(q)]}{\hat{\delta}^\top C^{-1} \hat{\delta}}, \quad q' = q. \quad (12)$$

Measurement pairs determine the correction direction; the preserved context sets the reference mean. The native step $\alpha = 1$ implements the score-level projection in Proposition 1 with fitted $w = C^{-1}\hat{\delta}$, metric $\Omega = C^{-1}$, and anchor $w^\top \hat{m}_Z(q)$. With an affine mean estimator, the resulting map is affine and requires no separately trained source critic.

Correction with learned source scores. Iterative MGPA fits source critics to the current representations. At stage k , critic j produces a source-logit difference $s_j^{(k)}(h)$, and a ridge-regression anchor estimates its conditional expectation:

$$a_j^{(k)}(q) \approx \mathbb{E}[s_j^{(k)}(H^{(k)}) \mid Q = q], \quad e_j^{(k)}(h) = s_j^{(k)}(h) - a_j^{(k)}(q(h)). \quad (13)$$

Anchors regress current FIT scores on the original FIT complement. Since $q(h + B\Delta z) = q(h)$, the fitted anchor remains constant during each edit. Suppressing the stage index, stack residuals in $e \in \mathbb{R}^J$ and define $A_j = \nabla_h s_j(h)^\top B$, so that $e(h + B\Delta z) \approx e(h) + A\Delta z$. A positive diagonal weighting W , based on gated gradient norms with a positive stabilizer, balances critic scales; write $\bar{A} = WA$ and $\bar{e} = We$. Within radius $\rho > 0$, the step jointly reduces the critics' linearized deviations from their conditional targets while penalizing movement:

$$\Delta z_* = \arg \min_{\|\Delta z\| \leq \rho} \left\{ \frac{1}{2} \|\bar{A}\Delta z + \bar{e}\|^2 + \frac{\tau}{2} \|\Delta z\|^2 \right\}, \quad h^+ = h + B\Delta z_*, \quad \tau > 0. \quad (14)$$

Linear and nonlinear critics provide fixed and observation-dependent gradients within the same gate. At the next stage, critics are refitted to corrected FIT representations to target remaining source structure; their new scores are regressed on the same original Q . Every stage, and hence the composition, preserves Q in exact arithmetic. The radius ρ controls the maximum displacement per stage; Appendix A.7 specifies the weights and solver.

Fit once, reuse. Adapter fitting uses calibration data and source labels, but no downstream task labels. Closed-form MGPA applies one fitted affine map; Iterative MGPA replays stored critics and anchors without refitting. Both apply without source identity or encoder updates. Correction strength is controlled by α or by a selected iterative prefix, optionally with a fractional final stage. Operating points may be fixed, matched by validation movement, or selected using development task labels; the selected map is then held fixed for reuse without target-task tuning. Algorithm 1 gives the iterative fitting and replay procedure, and Appendix B provides the numerical settings and endpoint rules.

6 EXPERIMENTS

We evaluate the MGPA construction in three steps: controlled measurements test selective correction and its components; recorded EEG tests source attenuation alongside task utility; cross-task reuse tests whether one selected correction can improve other frozen predictors. Closed-form and Iterative MGPA share the same principle and pooled conditional target. Their comparison examines complementary constructions, while targeted ablations isolate the gate, target and nonlinear critic.

6.1 EXPERIMENTAL DESIGN AND EVALUATION

Evaluation. Source accessibility and downstream utility answer different questions: information readable by a source probe need not be used by a task head. We report source AUROC S from the strongest independent reader in a declared bank, and task AUROCs for heads trained before correction (T_f , frozen) or after correction (T_r , refitted). These distinguish compatibility with an existing predictor from utility after head retraining. Task scores average the two cross-source transfer directions; recorded SSVEP uses twelve-class macro AUROC. Movement M is the participant-mean relative embedding displacement. The reuse study instead evaluates fixed heads across device-label associations and a separate control head trained without that association.

Fitting and comparison. Participant roles separate adapter fitting, validation, reader training and evaluation. MGPA fitting uses source labels and calibration contrasts, not downstream labels. Controlled gate comparisons and the supplementary N170 comparison with IGBP match validation movement budgets; main recorded comparisons use native outputs. Reuse selects an operating point on P300 development data, then holds it fixed on all evaluation tasks. Identity, LEACE and IGBP are source-blind comparators. CORAL and paired affine FEATMAP use source identity at application and are grouped separately. Iterative results average three fitted trajectories. Key contrasts use paired 95% participant-bootstrap intervals. Appendix B gives the splits, fitting repetitions, comparator settings, reader banks and uncertainty calculation.

6.2 CONTROLLED TESTS OF THE MGPA CONSTRUCTION

Selective correction from paired measurements. Using wet trials from the 102-participant SSVEP collection (Zhu et al., 2021), we encode each trial under common-average and Oz references. Full reference contrasts provide calibration; evaluated views use a scaled embedding contrast ($\gamma = .005$). Source assignment is associated with stimulus frequency during adapter fitting and balanced during evaluation. The source-task association makes source prediction an ambiguous guide to correction. Same-trial calibration supplies measurement evidence; the experiment tests whether MGPA can use it to attenuate source information while retaining the correlated task signal.

Closed-form MGPA brings source accessibility near chance while retaining frozen-task performance (Table 2). Relative to LEACE, the paired source change is $-.1496 [-.1580, -.1426]$ and the frozen-task gain is $.0964 [.0831, .1097]$. Routed CORAL and FEATMAP also attain near-chance source scores and retain task quality, using the known measurement condition.

Constraining critic-driven corrections. Do critic-driven updates benefit from measurement-derived rather than generic edit permissions? We compare measurement, same-rank PCA/random and ungated updates, holding the critic bank, solver, stage budget and original- Q_0 anchor input fixed; only measurement-gated edits preserve Q_0 . A separate target comparison holds the gate and remaining procedure fixed, contrasting conditional and empirical unconditional mean-score anchoring.

Table 2: **Selective correction with a single analytic map.** Controlled SSVEP at native endpoints: independent source AUROC, frozen/refitted task AUROC and relative movement. All methods share a reader realization over four rotations (81 distinct evaluation participants). Source-aware rows require source identity. Paired contrasts are in Appendix C.1.

Interface	Method	Source ↓	Frozen ↑	Refitted ↑	Movement
<i>Source-blind correction</i>	Identity	0.683	0.753	0.753	0.000
	LEACE	0.660	0.656	0.665	0.120
	MGPA-CF	0.510	0.753	0.753	0.028
<i>Source-aware alignment</i>	CORAL → ref. 0	0.502	0.753	0.753	0.012
	CORAL → ref. 1	0.502	0.753	0.753	0.012
	FEATMAP → ref. 0	0.502	0.753	0.753	0.012
	FEATMAP → ref. 1	0.502	0.753	0.753	0.012

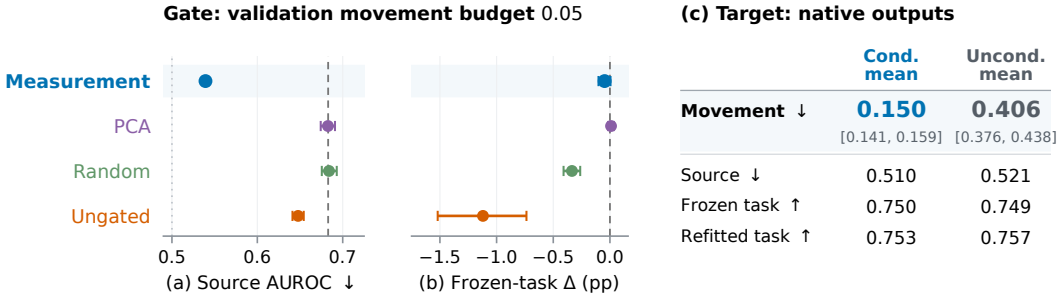


Figure 3: **Where correction acts and what it targets.** (a–b) *Gate*: source AUROC and paired frozen-task change from identity at a common .05 validation movement budget; dashed lines mark identity and the dotted line marks chance. Random averages five orientations. (c) *Target*: conditional versus empirical unconditional mean-score anchoring, with native 48-stage movement and source/task AUROC. Three fits, 81 participants; whiskers and movement brackets give 95% participant-bootstrap intervals. Budget-matched gates and native target endpoints are separate comparisons.

Measurement-gated correction suppresses source more strongly than all three controls (Figure 3a). Its frozen-task change from identity is $-.0005$ $[-.0010, .0001]$, and its advantage over ungated correction is $.0107$ $[.0069, .0147]$. Attained evaluation movements span $.0504$ – $.0539$. The result distinguishes measurement-supported directions from a generic restriction of rank or movement. Appendix B.2 details the comparison.

Conditioning reduces correction cost. Conditional-mean anchoring uses $M = .1500$ rather than $.4063$ for the empirical unconditional mean-score target, which requires about 2.7 times as much movement (Figure 3c). Source AUROC is lower ($.5105$ versus $.5214$), with no clear frozen-task difference; refitted-task AUROC favors the unconditional target ($.7574$ versus $.7527$). This links the cost argument in Section 4 to learned correction. Unlike the fixed-score analysis, these trajectories refit their critics separately. Appendix C.1 gives the target definitions and paired contrasts.

6.3 CORRECTION ON RECORDED EEG

We next test the construction on two forms of recorded EEG: 240-dimensional temporal N170 features from simultaneous Neuroscan/Flex recordings (Williams et al., 2020), and wet/dry SSVEP representations from frozen EEGPT. Table 3 compares source attenuation and task utility in both settings.

Temporal N170: nonlinear correction and task utility. Genuine calibration pairs support face/watch discrimination. The linear-only ablation keeps the measurement gate and anchor fixed while removing the nonlinear critic. At nearly identical movement ($.3449$ versus $.3469$), adding that

critic lowers source accessibility and improves refitted task performance. Relative to the linear-only ablation, the paired source AUROC change is $-.0413 [-.0647, -.0210]$ and the refitted-task gain is $.0171 [.0005, .0384]$. The full correction also improves both frozen and refitted task AUROC over identity (Table 3A; paired intervals in Appendix C).

Direct comparison with IGBP (Iskander et al., 2023) tests whether this benefit extends beyond our own critic ablation. Iterative MGPA improves native refitted-task AUROC over IGBP by $.0361 [.0306, .0405]$ with less movement ($.3469$ versus $.4973$). At a common $.25$ validation movement budget, it also achieves lower source AUROC and higher frozen/refitted task AUROC (Appendix C.2). FEATMAP toward Neuroscan attains stronger attenuation and higher task scores using source-aware routing at a larger displacement.

Recorded SSVEP: frozen-model representations. Actual wet/dry sessions test acquisition variation without constructed embedding interpolation. Because these sessions are not physiological trial pairs, calibration uses six FIT-local spectral and spatial transformations of individual trials. Both MGPA constructions correct the 512-dimensional token mean and add the same displacement to all tokens. Independent source readers and the twelve-frequency task access the complete 15×512 EEGPT representation.

Table 3: **Source attenuation and task utility on recorded EEG.** Native outputs: N170 (A, 10 participants) and wet/dry SSVEP (B, 21). S : maximum independent source AUROC over the eleven-reader bank in A and the 41-reader full-input bank in B; T_f/T_r : frozen/refitted task AUROC; M : relative movement. Iterative estimates average three fits; native movements differ. IGBP uses 100 projections with the same longer-training preset in both studies (Appendix B.2). Source-aware rows require device identity. Reference 0/1 denotes Neuroscan/Flex in A and dry/wet in B. N/A: paired FEATMAP requires physiological wet/dry trial pairs, unavailable in B.

Interface	Method	A. Temporal N170				B. Recorded SSVEP			
		$S \downarrow$	$T_f \uparrow$	$T_r \uparrow$	M	$S \downarrow$	$T_f \uparrow$	$T_r \uparrow$	M
<i>Source-blind correction</i>	Identity	0.989	0.728	0.728	0.000	0.916	0.688	0.688	0.000
	LEACE	0.980	0.728	0.727	0.234	0.918	0.689	0.689	0.102
	MGPA-CF	0.982	0.727	0.728	0.237	0.895	0.689	0.690	0.056
	MGPA-Iter	0.946	0.735	0.746	0.347	0.872	0.688	0.689	0.149
	Linear-only ablation	0.987	0.725	0.729	0.345	0.872	0.689	0.689	0.069
	IGBP	0.955	0.722	0.710	0.497	0.872	0.689	0.688	0.096
<i>Source-aware alignment</i>	CORAL $\rightarrow$ ref. 0	0.948	0.738	0.735	0.448	1.000	0.691	0.688	0.149
	CORAL $\rightarrow$ ref. 1	0.958	0.726	0.740	0.292	1.000	0.685	0.688	0.155
	FEATMAP $\rightarrow$ ref. 0	0.880	0.777	0.749	0.800	N/A	N/A	N/A	N/A
	FEATMAP $\rightarrow$ ref. 1	0.890	0.693	0.711	0.576	N/A	N/A	N/A	N/A

Both constructions reduce full-input source accessibility relative to LEACE (Table 3B). The paired source differences for Closed-form and Iterative MGPA are $-.0231 [-.0459, -.0058]$ and $-.0463 [-.0671, -.0225]$. Frozen/refitted task scores remain close to identity for both constructions. Thus attenuation extends to the full downstream input, with little observed change in task utility. IGBP reaches the same full-input source score ($S = .872$) with close task AUROCs and less movement ($.096$ versus $.149$). Appendix C.3 details this shared-interface comparison. Appendix B.3 specifies the full-input audit, and Appendix C reports paired task and component comparisons.

6.4 SELECTING ON P300, REUSING ON N170 AND MMN

The preceding studies assess what remains accessible in corrected representations. We now ask whether one correction can improve the behavior of predictors for other tasks. Simultaneous Neuroscan/Flex recordings of P300, N170 and MMN share a fixed EEGPT/PCA representation. We fit and select each adapter on P300, then apply the same map to N170 and MMN without target-task adapter tuning. The encoder and task heads remain frozen, so the comparison tests whether the reused correction improves existing predictions rather than enabling a newly trained predictor.

For each task, a head trained with device-label association is evaluated as that association weakens or reverses. Association is changed by selecting the observed device from real event pairs, without

altering the EEG signal. An independently trained control head measures utility without the induced association. P300 development data select correction strength under a worst-association objective and task-utility guardrails, with the same selection criterion for competitors. Appendix B.4 specifies the fitting and selection protocol.

Table 4: **One P300-selected correction, three frozen task heads.** Eight common participants; worst-association AUROC of shortcut-trained heads and independent-regime AUROC of separate control heads. Iterative scores average three fits, each with an unchanged P300-selected map. For each fit, worst is the minimum participant-mean AUROC over association strength. Paired intervals and per-fit results are in Appendix C.4.

Interface	Method	Worst-association AUROC $\uparrow$			Control-head AUROC $\uparrow$		
		P300	N170	MMN	P300	N170	MMN
<i>Source-blind correction</i>	Identity	0.355	0.122	0.446	0.627	0.622	0.640
	LEACE	0.601	0.384	0.614	0.622	0.621	0.639
	MGPA-CF	0.599	0.420	0.642	0.627	0.621	0.642
	MGPA-Iter	0.599	0.441	0.632	0.627	0.621	0.641
<i>Source-aware alignment</i>	CORAL	0.550	0.277	0.624	0.627	0.622	0.641
	FEATMAP	0.573	0.471	0.618	0.627	0.611	0.625

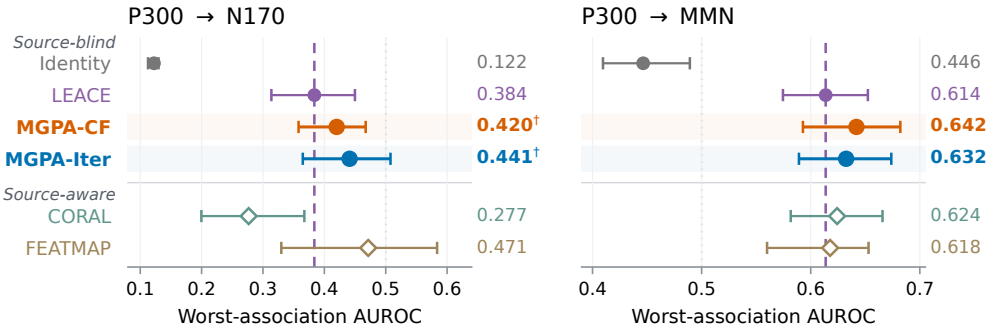


Figure 4: **Reusing a P300-selected correction.** Absolute worst-association AUROC of frozen N170/MMN shortcut-trained heads; whiskers show marginal 95% participant-bootstrap intervals. Dashed: LEACE; dotted: chance; hollow: source-aware methods. $\dagger$: paired 95% interval for the gain over LEACE entirely above zero. Intervals condition on fitted maps and are unadjusted for multiplicity; panel ranges differ.

Both MGPA constructions improve worst-association AUROC over LEACE on the two transfer tasks (Table 4). Iterative MGPA gains .0574 [.0338, .0831] on N170 and .0188 [.0040, .0241] on MMN; Closed-form MGPA gains .0364 [.0049, .0667] and .0282 [.0021, .0343], respectively. Control-head AUROC stays close to identity across all three tasks. FEATMAP has the highest N170 worst-association score, with lower N170 and MMN control-head scores; the grouped comparison retains its source-aware deployment interface.

Figure 4 compares absolute worst-association performance on both transfer tasks, with LEACE as a reference. Together, the experiments connect selective correction of representations to reusable changes in downstream prediction.

7 DISCUSSION AND CONCLUSION

MGPA provides a practical way to correct acquisition-sensitive variation in frozen representations without treating all source-predictive variation as disposable. Calibration specifies where edits are supported; it does not imply that everything within that space should be removed. The correction target must also account for what is preserved: editable coordinates should not be forced to compensate for source information that remains available in the fixed complement.

This perspective opens a practical route to adapting pretrained representations without rebuilding the encoder for each downstream task. The value of a reusable correction lies not only in changing what a probe can read, but in changing how an existing predictor behaves when acquisition cues cease to be reliable. Closed-form and Iterative MGPAs provide two implementations of this principle. Although evaluated here on EEG, the construction is not EEG-specific: it requires credible calibration views of the same observation under different measurement conditions. Richer calibration could reveal which edit directions remain supported across acquisition conditions, and which depend on a particular contrast. The opportunity is to make representation correction more selective and its assumptions more explicit: *measurement evidence determines where correction may act, and preserved information shapes what it should aim for.*

REPRODUCIBILITY STATEMENT

The supplement provides proofs, calibration and update specifications, participant roles, selection rules and statistical evaluation details. Code, configurations and participant-level results are indexed by the repository README (Appendix D), which distinguishes saved-result reconstruction from fresh experiments.

ETHICS STATEMENT

This study uses existing research EEG datasets and introduces no new data collection or clinical intervention. Provenance attenuation does not confer anonymity or establish clinical validity. Applications involving patients require task-specific validation and appropriate data governance.

AI USAGE STATEMENT

Generative artificial intelligence (AI) tools assisted code cleaning, preliminary literature review and manuscript polishing. The authors retain responsibility for the final content.

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A PROOFS AND SOLVER DETAILS

All statements in this section condition on the fitted gate and treat its orthonormal basis B as fixed. Fixed-score statements also condition on the critic and correction metric. Norms without a subscript are Euclidean; $\|v\|_\Omega^2 = v^\top \Omega v$.

A.1 FIXED-COMPLEMENT PRESERVATION AND THE SOURCE FLOOR

Since $(I - P)B = 0$, every admissible correction satisfies

$$(I - P)R(h) = (I - P)h = q(h).$$

Thus Q is recoverable from the corrected representation, and the chain rule gives (2). Preservation also holds under composition and fractional steps. For $R_\mu(h) = q(h) + B\mu(q(h))$, the output is determined by Q , so the conditional-information term is zero.

A.2 FIXED-SCORE GEOMETRY AND RETAINED INFORMATION

In Proposition 1, the functions $b(q)$, $w(q)$ and $\Omega(q) \succ 0$ are measurable in q , with $w(q) \neq 0$ almost surely. The score has finite conditional second moments when costs are evaluated. For fixed (q, z) , matching anchor $a(q)$ requires $w^\top \Delta = a(q) - s(q, z)$. Metric Cauchy–Schwarz gives

$$(a - s)^2 \leq (w^\top \Omega^{-1} w) \Delta^\top \Omega \Delta.$$

The unique minimizer is $\Delta = v(a - s)$, where $v = \Omega^{-1}w/d_\Omega$. Its squared norm is $(a - s)^2/d_\Omega$. The output editable block is

$$Z' = N + v\{a(Q) - b(Q)\}, \quad N = (I - vw^\top)Z.$$

The output is a measurable function of (Q, N) ; conversely, $N = Z' - v\{a(Q) - b(Q)\}$ is measurable in the output because Q is retained. Thus $\sigma(T_a) = \sigma(Q, N)$. Every anchor retains the same information about U and about any task variable Y , including the same unrestricted Bayes prediction risk. Conditional independence from U holds exactly when $N \perp U \mid Q$. The conditional bias–variance decomposition of $\mathbb{E}[(s - a)^2 \mid Q = q]$ proves (6) and the unique mean-anchor optimum.

For $\Omega = I$, the edit is $w(a - s)/\|w\|^2$ and the denominator is $\|w\|^2$. Orthonormality of B makes this the Euclidean movement in the standardized representation as well. The result compares exact score-level corrections with the same fixed score and metric. Nonlinear sample-dependent Jacobians, active radius constraints and independently refitted compositions are not members of that comparison.

A.3 PROOF OF THEOREM 2

Fix q and abbreviate $p = p(q)$, $m = m(q)$ and $\kappa = \kappa(q)$. Expanding the two Gaussian quadratic log densities gives

$$\log \frac{p(z \mid q, 1)}{p(z \mid q, 0)} = \delta^\top S^{-1}(z - m) = \ell(q, z).$$

Let $L = I - \delta\delta^\top S^{-1}/D$. The repaired editable coordinate is

$$Z_\kappa = LZ + \frac{\delta}{D}\{\delta^\top S^{-1}m + \kappa\}.$$

Since $L\delta = 0$, its conditional mean is the same for both sources. Its conditional covariance is $LSL^\top$ for both. The resulting possibly singular Gaussian laws are identical, proving conditional independence. The map preserves Q by construction.

Proposition 1, applied with $w = S^{-1}\delta$ and $\Omega = S^{-1}$, gives the unique minimum-norm edit $\Delta = \delta(\kappa - \ell)/D$, with squared cost $(\kappa - \ell)^2/D$. Conditional on $Q = q$, $U = u$, the residual score ℓ has mean $(2u - 1)D/2$ and variance D . Consequently,

$$\mathbb{E}[\ell \mid q] = (2p - 1)D/2, \quad \text{Var}(\ell \mid q) = D + D^2p(1 - p).$$

Thus

$$C_\kappa(q) = \frac{\mathbb{E}[(\ell - \kappa)^2 | q]}{D} = 1 + Dp(1-p) + \frac{\{\kappa - (2p-1)D/2\}^2}{D},$$

which is (9). Its minimum occurs at $\kappa_* = (2p-1)D/2$. Subtracting this minimum from the cost at $\kappa = -s_Q$ yields (10). Since $s_Q = \text{logit } p$ and $2p-1$ have the same sign, the excess is strictly positive for $p \neq 1/2$. At fixed such p , its numerator tends to $s_Q^2 > 0$ as $D \downarrow 0$, so the excess diverges.

A.4 SELECTIVE MEAN ANCHORING AND FULL COMPLETION

The unique minimizer of (9) is $\kappa_* = (2p-1)D/2$. To show its optimality among deterministic conditional erasers affine in Z , whiten at fixed q :

$$X = S^{-1/2}(Z - m), \quad v = S^{-1/2}\delta, \quad X | U = u \sim \mathcal{N}((2u-1)v/2, I), \quad \|v\|^2 = D.$$

An affine map $X' = FX + c$ erases source exactly if and only if $Fv = 0$: the two output means must coincide, and their covariance already agrees. For fixed F , the least-distortion intercept is $c = (I - F)\mathbb{E}X$. Since $\text{Cov}(X) = I + p(1-p)vv^\top$, its cost under the erasure constraint is

$$\text{tr}\{(F - I)\text{Cov}(X)(F - I)^\top\} = \|F - I\|_F^2 + p(1-p)D, \quad Fv = 0.$$

The unique matrix nearest I in Frobenius norm subject to $Fv = 0$ is $F_* = I - vv^\top/D$. Unwhitening gives precisely R_{κ_*} , with cost $1 + Dp(1-p)$. This is the conditional binary affine-erasure geometry of LEACE (Belrose et al., 2023).

For quotient-only completion, the best fixed- q choice is $Z' = \mathbb{E}[Z | q]$. Its cost is the S^{-1} trace of $\text{Cov}(Z | q) = S + p(1-p)\delta\delta^\top$, hence $r + Dp(1-p)$, exceeding the selective correction cost by $r - 1$.

A.5 PROOF OF COROLLARY 3

Fix the fitted quantities and q . The editable part of T_a is $Z' = MZ + v\{a(q) - b(q)\}$, where

$$M = I - \frac{\widehat{\delta\delta^\top}C^{-1}}{\widehat{\delta^\top}C^{-1}\widehat{\delta}}, \quad \ker M = \text{span}(\widehat{\delta}).$$

The translation is shared by both sources. Their output mean difference is $M\delta$ and their common covariance is $MSM^\top$. With $+$ denoting the Moore–Penrose inverse, put $A = MS^{1/2}$ and $x = S^{-1/2}\delta$. An SVD shows that $A^\top(AA^\top)^+A$ is the orthogonal projector onto $(\ker A)^\perp$. Since $\ker A = \text{span}(S^{-1/2}\widehat{\delta})$,

$$D_{\text{out}} = (M\delta)^\top(MSM^\top)^+(M\delta) = x^\top A^\top(AA^\top)^+Ax = D - \frac{(\delta^\top S^{-1}\widehat{\delta})^2}{\widehat{\delta^\top}S^{-1}\widehat{\delta}}.$$

Dividing by $D > 0$ proves (11). No equality between the fitted C and evaluation covariance S is needed. The latter defines the angle and separation; it need not be available to the algorithm.

This is a conditional identity for any fitted direction, not a bound on its estimation error. Such a bound requires a calibration model relating the paired contrasts to the conditional shift δ ; the learned gate alone does not supply it. The full-step qualification is essential: for $M_\alpha = I - \alpha(I - M)$, $\det M_\alpha = 1 - \alpha$. Thus for $\alpha \neq 1$ the map is invertible given Q and preserves the exact conditional source information and covariance-adjusted separation, even though a fixed prediction head can respond differently.

A.6 ESTIMATING THE SAME CONDITIONAL-MEAN ANCHOR

Fix the gate, affine score, metric and evaluation distribution in Proposition 1. Let an anchor-fitting sample $\mathcal{C}$, independent of the evaluation observation, produce a Q -dependent anchor $\widehat{a}_\mathcal{C}(q)$ with finite mean-squared error. Writing $\mu(q) = \mathbb{E}[s | Q = q]$, Proposition 1 and expectation over $\mathcal{C}$ give

$$\mathbb{E}_\mathcal{C} \mathcal{C}_{\widehat{a}_\mathcal{C}}(q) - \mathcal{C}_\mu(q) = \frac{\mathbb{E}_\mathcal{C} \{\widehat{a}_\mathcal{C}(q) - \mu(q)\}^2}{d_\Omega(q)}. \quad (15)$$

Thus the estimated anchor has lower expected squared movement than a fixed reference $a_0(q)$ exactly when its estimation mean-squared error is below $\{a_0(q) - \mu(q)\}^2$. This is an out-of-sample comparison conditional on the fixed score and gate.

A.7 THE SMOOTH TRUST-REGION STEP

For the editable Jacobian A and critic residual vector e in Section 5, smooth weighting balances the gated gradient scales:

$$W_{jj} = (\|A_j\|^2 + \lambda^2)^{-1/2}, \quad \bar{A} = WA, \quad \bar{e} = We, \quad \lambda > 0. \quad (16)$$

For $\lambda, \tau, \rho > 0$, the objective in (14) is strictly convex on a compact convex ball. Its unique minimizer has the standard trust-region form (Moré & Sorensen, 1983)

$$\Delta z_* = -\bar{A}^\top \{\bar{A}\bar{A}^\top + (\tau + \nu)I\}^{-1} \bar{e}, \quad \nu \geq 0, \quad \nu(\|\Delta z_*\| - \rho) = 0. \quad (17)$$

The stationarity condition

$$(\bar{A}^\top \bar{A} + (\tau + \nu)I)\Delta z_* = -\bar{A}^\top \bar{e}$$

yields the stated $J \times J$ solution. Set $\nu = 0$ when the unconstrained step is feasible; otherwise its norm decreases continuously with ν , and a scalar solve enforces $\|\Delta z_*\| = \rho$. Since B is orthonormal, the radius directly bounds the embedding edit: $\|h^+ - h\| = \|\Delta z_*\| \leq \rho$.

B EXPERIMENTAL PROTOCOLS

FIT determines coordinates, gates and adapters; VAL sets prescribed movement endpoints; HEAD trains independent readers; EVAL supplies participant scores. Participant roles are disjoint within a split. Table 5 distinguishes native outputs, matched movement and task-informed P300 DEV selection. Downstream labels construct associations and train heads, but enter adapter selection only in reuse and never adapter fitting. The repository README (Appendix D) indexes participant lists, executable configurations and per-study reproduction instructions.

Table 5: **Protocol map.** Participant counts are FIT/VAL/HEAD/EVAL (DEV replaces VAL in reuse). Gates are fixed during correction.

Study	Coordinates, calibration and gate	Participants and operating point
Controlled SSVEP	Pooled EEGPT512; full same-trial CAR/Oz contrasts; 90% directional energy, ranks 3, 3, 3, 2	Four role rotations; 81 distinct EVAL participants. Native correction/target; .05 VAL budget for permission.
Temporal N170	240 temporal features; simultaneous Neuroscan/Flex pairs; 90% energy, rank 56	4/2/4/10; native maps and matched-budget IGBP comparison.
Recorded SSVEP	Mean512 correction, 15×512 readouts; six virtual FIT-local contrasts; 99% energy, rank 328	41/20/20/21; native maps.
P300-selected reuse	EEGPT/PCA256, P300 FIT reference coordinates; 400 simultaneous CAL pairs, rank 8	4/2/4/8; P300 DEV-selected map, unchanged on N170/MMN.

B.1 MEASUREMENT VIEWS AND PARTICIPANT ROLES

Controlled SSVEP. The wet/dry collection contains 102 participants, 12 frequencies and ten blocks per frequency and electrode type (Zhu et al., 2021). Wet trials from eight occipital channels are cropped to samples 160–659 at 250 Hz, standardized within channel and encoded by frozen EEGPT (Wang et al., 2024). CAR and Oz-reference views are encoded separately; averaging 15 tokens yields $h_i^{\text{CAR}}, h_i^{\text{Oz}} \in \mathbb{R}^{512}$. The evaluated endpoints are

$$h_i(u) = \frac{1}{2}(h_i^{\text{CAR}} + h_i^{\text{Oz}}) + (2u - 1)\frac{\gamma}{2}(h_i^{\text{Oz}} - h_i^{\text{CAR}}), \quad \gamma = .005,$$

whereas calibration uses the full CAR–Oz contrasts. The task is frequency ≥ 12 Hz. Within each FIT/VAL participant–frequency cell, nine of ten trials receive source 1 in the high-frequency class and one of ten in the low-frequency class. Adapters fit these observed endpoints; HEAD/EVAL readers use both endpoints with equal source mass.

Five folds are stratified by electrode-session order and gender. In rotation r , folds $r, r+1, r+2$ supply EVAL, HEAD, VAL cyclically, and the remaining two supply FIT. The four FIT/VAL/HEAD/EVAL counts are 41/20/20/21, 42/20/20/20, 41/21/20/20 and 40/21/21/20. Each participant supplies 120 underlying trials. The 81 distinct EVAL participants appear once per fit seed; source assignment is fixed across seeds. Closed-form comparisons use one deterministic map per split and the seed-17 reader realization; gate/target comparisons cross these rotations with all shared fit seeds.

Temporal N170. Neuroscan (source 0) and Flex (source 1) record the same watch/face events concurrently (Williams et al., 2020). Processing uses 16 common channels, 0.1–30 Hz zero-phase FIR filtering, CAR, resampling to 128 Hz, and one-second epochs starting at $-.2$ s with prestimulus baseline subtraction. Event ordinals and timestamps align the views. A pair is retained only if neither device exceeds $100 \mu\text{V}$ absolute amplitude after baseline correction. FIT/VAL/HEAD/EVAL contain 622/288/716/1381 clean pairs; every EVAL participant retains both classes. Fifteen successive 50 ms channel averages from 0 to 750 ms give 240 coordinates without an encoder or PCA. This is distinct from the EEGPT-based reuse protocol below.

Recorded SSVEP. Source 0/1 denotes actual dry/wet sessions, with no physiological trial correspondence. The roles contain 9840/4800/4800/5040 observations under one fixed stratified split. Six calibration transformations of each trial use spectral transport, spatial transport and their composition at strengths .5 and 1 toward the opposite source. FIT prestimulus signals determine smoothed log-power gains clipped to $[-.25, 4]$ and spatial whitening/coloring with covariance shrinkage .1. MGPA corrects the token mean and adds its displacement to every token; source and twelve-frequency task readers access the full 15×512 representation.

B.2 FITTING, REPLAY AND COMPARISON INTERFACES

Table 6: **Shared MGPA settings.** Component ablations change only the specified critic or target. Split, reader and assignment seeds are separate.

Component	Setting
Iterative fits / horizon	Seeds 17, 29, 43; at most 48 stages
Source critics	Balanced logistic ($C = 1$); GELU MLP, hidden width 128
Conditional-mean anchor	Ridge with intercept and penalty 10 (both constructions)
Neural-critic optimizer	AdamW, batch 256, weight decay .001
Gradient stabilizer / damping	.1 / .001
Controlled and real-representation fits	12 critic epochs, LR .001, radius $.05\sqrt{d}$
P300 reuse fits	At most 100 epochs, LR .0003, source-VAL patience 10, radius $.025\sqrt{d}$
Closed-form residual covariance	.01 shrinkage toward mean-diagonal identity
Participant uncertainty	20,000 percentile bootstrap resamples, seed 1101

Algorithm 1 Iterative MGPA: fit once, replay without source labels

- Inputs:** calibration pairs, source-labeled FIT/VAL embeddings, gate rank or energy threshold, critic bank, λ, τ, ρ, K , and endpoint rule. Task-informed selection additionally requires DEV task data.
- Fit standardization and the contrast-SVD gate (3); freeze B, P, Q . Return identity if $r = 0$. Store original FIT/VAL complements.
- Fit source critics on current FIT embeddings and regress their scores on original FIT Q to obtain conditional anchors.
- Store critics and anchors; compute e, A, W , solve (14), and apply $h \leftarrow h + B\Delta z_*$. Record VAL outputs and repeat Steps 3–4 up to K stages.
- Select and store the prefix and optional final-stage fraction using the declared validation rule; only task-informed selection uses task labels.
- Deployment:** standardize one embedding, replay the stored map, and return it in the downstream interface’s coordinates.

Standardization uses FIT only. Gates use row-normalized contrasts; controlled and N170 fits filter contrast norms at the fifth percentile, whereas recorded SSVEP uses no percentile filter. Controlled gating uses at most 2048 deterministically sampled contrasts. Energy thresholds and ranks appear in Table 5. Closed-form MGPA uses the same gate and standardization, the mean projected calibration contrast, and the shared Ridge anchor. Its metric comes from residuals of a joint regression of Z on Q and source: intercept/source coefficients are unpenalized and Q has penalty 10. The residual second moment is normalized by FIT count and shrunk as in Table 6; (12) uses linear solves. Native studies take $\alpha = 1$; reuse selects α on DEV. Recorded contrasts are oriented wet-like to dry-like for both gate and direction.

The target ablation replaces the conditional anchor by the current-stage empirical FIT-mean score; a zero-score control is also reported. The linear-only ablation removes the nonlinear critic. Other fitting settings remain matched as specified in the corresponding results.

Baselines and deployment information. LEACE fits an unrestricted covariance-weighted affine eraser to observed FIT rows. CORAL uses FIT source means and identity-regularized sample covariances, not pair correspondence. FEATMAP fits centered unregularized least squares to FIT pairs, with the minimum-norm slope for rank-deficient designs; controlled pairs use the evaluated γ -scaled endpoints. CORAL/FEATMAP leave their chosen reference unchanged and require source routing at deployment; both orientations are reported. MGPA, LEACE and IGBP deploy source-blind. Recorded SSVEP has no paired FEATMAP comparison. No method uses EVAL moments. Reuse searches regularization/strength on P300 DEV instead of using these native settings.

IGBP. IGBP (Iskander et al., 2023) shares each study’s roles, standardized inputs and independent readers. At each iteration a $d \rightarrow d \rightarrow 2$ ReLU classifier drives an ungated projective update without a conditional anchor or pair correspondence. All reported fits use the longer-training preset: AdamW, LR 2×10^{-4} , weight decay .01, batch 256, at most 50 epochs, VAL source-accuracy patience 10 and improvement threshold .002, with no .80 accuracy exit. Native maps apply 100 projections without task-based stopping. Updates are evaluated directly in logit space to avoid softmax saturation. For N170, $d = 240$; for recorded SSVEP, $d = 512$ and the resulting mean correction is added to every token, matching MGPA’s interface.

Matched movement and edit-permission controls. Native outputs apply the complete fitted map; movements need not match. A budget b instead selects the first upward crossing of $\|H'_{\text{VAL}} - H_{\text{VAL}}\|_F / \|H_{\text{VAL}}\|_F = b$, interpolating only the last update and replaying that prefix/fraction on HEAD/EVAL. Controlled permission compares the measurement gate, leading observed-FIT PCA directions of the same rank, five fixed random orientations of that rank (averaged, not selected), and all 512 directions. Each arm refits its critics and anchors with the same recipe and original measurement complement Q_0 as anchor input. Thus controls receive calibration-derived rank and conditioning information; only the measurement arm necessarily preserves Q_0 . Every correction arm and random orientation attains the reported .05 budget; all 108 cells, including identity, are available. At the originally primary .10 budget, 17 of 60 random trajectories do not attain it within the horizon; native endpoints are not substituted. The N170 IGBP budget comparison uses the same first-crossing rule.

B.3 INDEPENDENT READERS AND STATISTICAL SUMMARIES

Source readers fit corrected HEAD outputs independently of adapters and score participant-disjoint EVAL outputs. Table 7 specifies their inputs and capacity. All preprocessing uses HEAD only; each recording, not each token, is one example. Neural readers train without early stopping and contribute three restarts at 12/50/200 epochs.

Recorded common-offset correction preserves $R_t = H_t - \bar{H}$. Residual readers fit once on original HEAD residuals and are reused on corrected EVAL; numerical invariance is checked. This retained token variation is distinct from the pooled gate complement Q .

Task utility and movement. Controlled/N170 heads are HEAD-standardized balanced Ridge classifiers (penalty 10, LSQR). The recorded task head uses HEAD-standardized flattened features and the token architecture above with a twelve-class output, trained for 50 epochs. Frozen heads and

Table 7: **Independent source-reader bank.** Controlled/N170 use the 11 common readers. Recorded SSVEP uses all $11 + 20 + 10 = 41$ readers; its common bank alone defines S_{mean} .

Input	Readers and training
Common representation	Ordinary and covariance-whitened logistic ($C = 1$, whitening shrinkage .01); GELU MLP with width 128, AdamW, LR .001, batch 256, weight decay .001.
Direct full tokens	Raw logistic ($C = 1$), HEAD-standardized logistic ($C = .01$); flat $7680 \rightarrow 128 \rightarrow 2$ GELU with common training; token network with LayerNorm, shared $512 \rightarrow 64$ GELU, concatenation and binary output, LR .0005, batch 128, weight decay .001, gradient clipping 5.
Centered token residuals	Standardized logistic and the same token network with its three restarts/checkpoints.

scalers train on original HEAD; refitted heads and scalers train on corrected HEAD; both evaluate corrected EVAL. T_f/T_r equally average participant AUROCs over $0 \rightarrow 1$ and $1 \rightarrow 0$ transfer; recorded scores are twelve-class macro AUROC. M averages $\|H'_p - H_p\|_F / \|H_p\|_F$ over EVAL participants, which can differ from aggregate VAL movement.

Aggregation and uncertainty. Source accessibility maximizes participant-mean AUROC over readers within each fit, then averages fits. Controlled scores pool disjoint EVAL folds; deterministic maps contribute one output per split. Shared participant bootstrap draws pair methods, transfer directions and fits, recomputing the source maximum inside every draw. Intervals condition on fitted coordinates, adapters and readers and are pointwise, without multiplicity adjustment; seeds/trials do not increase participant counts. These cohorts informed protocol development, including the P300 selection task and rank eight. Controlled gate/target studies share participants; the two N170 protocols overlap. Source maxima measure accessibility under the specified bank, not complete erasure or unbiased performance of an EVAL-selected reader.

B.4 P300-SELECTED CORRECTION REUSED ON N170 AND MMN

Calibration and shared coordinates. All three tasks use simultaneous Neuroscan/Flex recordings and the same eight EVAL participants. The first 100 consecutive P300 pairs per FIT participant form CAL. Any other FIT event whose one-second epoch overlaps CAL on either device is excluded, leaving 2260 observed events. One endpoint per observed event plus both endpoints of the 400 CAL pairs yields 3060 unique fitting endpoints for MGPA, LEACE and CORAL; FEATMAP uses the 400 pairs. CAL supplies both MGPA gates and the Closed-form direction; Iterative critics fit the pooled bank without paired targets.

Frozen EEGPT retains $7 \times 4 \times 512 = 14,336$ features from seven overlapping windows, with fixed $20 \mu\text{V}$ input scaling. Centered PCA256 and standardization fit only the Neuroscan endpoints of remaining P300 FIT events, then apply to both devices and all tasks. Neither N170 nor MMN fits coordinates or selects the adapter. HEAD uses all technically valid events, unlike the amplitude-filtered temporal N170 study.

Frozen heads and association shifts. Each task has two balanced logistic heads trained on original HEAD features with equal participant weights: an aligned, shortcut-sensitive head and an independent-assignment control head. Regularization is selected from $C \in \{10^{-6}, 10^{-5}, \dots, 10\}$ by leave-one-HEAD-participant-out independent-head AUROC and shared between that task’s heads. Heads, standardizers and assignment realizations stay fixed across adapter fits. For positive-class prevalence π_p , endpoint assignment satisfies

$$\Pr(U = 1 \mid Y = y, p; \xi) = \frac{1}{2} + .4\xi \frac{y - \pi_p}{\max(\pi_p, 1 - \pi_p)}, \quad \xi \in [-1, 1].$$

Aligned/independent/reversed settings are $\xi = 1/0/-1$; device marginals remain one half and signals are unchanged. Fitting samples recorded endpoints; evaluation weights both endpoints. Worst AUROC is the exact minimum of the participant-mean quadratic AUROC curve on $[-1, 1]$, including interior minima. Control-head AUROC is evaluated at $\xi = 0$.

Selection and transfer. Both MGPA constructions use rank-eight gates and pooled conditional-mean targets. Iterative training follows Table 6, with 1332 DEV source-validation endpoints. P300 DEV selects the endpoint maximizing worst AUROC subject to at most .01 loss from identity for *each* head at independent association. Ties prefer less movement, then an earlier endpoint. All iterative prefixes are eligible. Affine strengths are 0, .02, . . . , 1, 1.1, 1.2, 1.3; baseline ridge grids and both CORAL/FEATMAP reference choices are searched under the same rule. Across recipes within a family, candidates within .002 of the best admissible DEV worst AUROC are ordered by movement and the fixed tie rule. The grids and selected values are recorded in the repository’s task-reuse guide; Closed-form MGPA selects $\alpha = 1.1$. Each complete selected map is then reused unchanged on P300, N170 and MMN.

Reuse movement for DEV tie-breaking is the participant mean of the ratio of mean paired-event displacement norm to mean input norm, rather than the Frobenius ratio above. Reported worst AUROC averages fit-specific minima of participant-mean curves, not the minimum of an ensemble curve. Each shared participant-bootstrap draw recomputes every curve and its minimum before averaging fits; the same draws pair all tasks and methods. Control-head AUROCs average participants and fits. Other uncertainty conventions follow Appendix B.3.

C SUPPORTING EXPERIMENTAL RESULTS

Paired intervals follow Appendix B.3.

C.1 CONTROLLED CORRECTION AND TARGET CHOICE

Closed-form correction. In Table 2, Closed-form MGPA changes source AUROC from identity by $-.173 [-.180, -.164]$. Frozen- and refitted-task changes are $-.0002 [-.0007, +.0002]$ and $+.0004 [-.0001, +.0008]$; the refitted-task gain over LEACE is $+.0879 [.0762, .0998]$.

Conditional versus unconditional targets. The unconditional control uses each critic’s empirical mean score on the current corrected FIT inputs, recomputed at every stage without downstream labels or HEAD/EVAL data. Gate, critic fitting, solver and the 48-stage horizon are unchanged. Each trajectory refits its critics, testing learned correction rather than the fixed-score equivalence of Proposition 1. Zero-score targeting remains a separate control: balanced source labels do not imply a zero mean logit.

Table 8: **Conditional versus constant targets.** Native controlled outputs for the same 81 participants and fits as Figure 3c. S : source AUROC; T_f/T_r : frozen/refitted task AUROC; M : movement.

Target	$S \downarrow$	$T_f \uparrow$	$T_r \uparrow$	M
Conditional mean	0.5105	0.7496	0.7527	0.1500
Unconditional mean (FIT)	0.5214	0.7488	0.7574	0.4063
Zero score	0.5199	0.7479	0.7562	0.4172

Conditional minus empirical-unconditional anchoring gives movement $-.2563 [-.2907, -.2232]$ and source AUROC $-.0110 [-.0143, -.0079]$. Frozen-task change is $+.0008 [-.0015, .0030]$; refitted-task change is $-.0047 [-.0070, -.0025]$.

C.2 TEMPORAL N170: NONLINEAR CORRECTION AND TASK UTILITY

Linear-only ablation. Removing the GELU critic retains the gate, Ridge mean anchor, smooth weighting, trust-region radius and stage horizon. Single-critic damping .0005 preserves regularization relative to mean critic loss, equivalently duplicating the affine constraint at the shared two-critic damping. Sample-dependent bounded steps make this a linear-critic ablation, not a globally affine map. Closed-form MGPA instead estimates its direction and covariance metric from measurement contrasts and the joint residual model. Table 9 shows why the distinction matters: linear-only correction lowers linear accessibility but leaves nonlinear accessibility nearly intact. Iterative MGPA lowers the full-bank score with higher task scores at nearly the same movement.

Table 9: **Nonlinear critics on N170.** A: strongest linear reader L versus full-bank S for the ten participants in Table 3A. B: Iterative minus linear-only AUROC percentage points, with paired 95% intervals. Iterative estimates average fits; linear-only is deterministic.

A. Native source accessibility			
Correction	$L \downarrow$		$S \downarrow$
Identity	0.763		0.989
MGPA-Iter	0.794		0.946
Linear-only	0.663		0.987
B. MGPA-Iter minus Linear-only (AUROC percentage points)			
Contrast	$\Delta S \downarrow$ [95% CI]	$\Delta T_f \uparrow$ [95% CI]	$\Delta T_r \uparrow$ [95% CI]
Paired difference	-4.13 [-6.47, -2.10]	+0.93 [-0.05, +1.88]	+1.71 [+0.05, +3.84]

Comparison with IGBP. At native outputs, Iterative MGPA minus IGBP gives frozen/refitted task gains of $+0.0123$ [.0042, .0203] and $+0.0361$ [.0306, .0405], and movement lower by .1504 [.1230, .1769]. The source-AUROC difference is -0.0091 [-.0198, .0006]. At common VAL movement budget .25 (Table 10), the respective source, frozen-task and refitted-task differences are -0.0234 [-.0393, -.0099], $+0.0082$ [.0015, .0154] and $+0.0156$ [.0063, .0269]. At budget .10, source accessibility is lower but a downstream advantage is not established. These comparisons assess complete correction procedures; equal iteration counts would not equalize their computation.

Table 10: **N170: MGPA versus IGBP at matched VAL budgets.** First-crossing endpoints follow Appendix B.2; attained EVAL movement M may differ. Same ten participants and shared fits.

VAL budget	Method	$S \downarrow$	$T_f \uparrow$	$T_r \uparrow$	M
.10	MGPA-Iter	0.9850	0.7274	0.7300	0.0974
.10	IGBP	0.9893	0.7292	0.7284	0.0898
.25	MGPA-Iter	0.9643	0.7292	0.7394	0.2422
.25	IGBP	0.9877	0.7210	0.7237	0.2239

Task utility. Iterative MGPA improves frozen/refitted task AUROC over identity by $+0.0070$ [$+0.0012$, $+0.0131$] and $+0.0187$ [$+0.0042$, $+0.0366$], respectively. Both transfer directions receive equal weight: the refitted gain is concentrated in Neuroscan-to-Flex (.608 to .648), while the reverse direction changes from .847 to .844.

C.3 RECORDED SSVEP: FULL-INPUT ACCESSIBILITY AND UTILITY

On the same saved outputs and participant roles, Iterative MGPA gives pooled source AUROC .7954 and full-bank $S = .8718$, versus identity’s .9155. Readers trained directly on all tokens independently support attenuation: their maximum falls from .8987 to .8685, a paired change of -0.0302 [-.0459, -.0174]. The full-bank difference from LEACE is -0.0463 [-.0671, -.0225]. Closed-form MGPA gives $S = .8949$, a change of -0.0206 [-.0356, -.0043] from identity. All maps share the 41-reader bank (Appendix B.3).

Residual-only readers attain .8718 for every common-offset correction; this determines the full-bank maximum for Iterative MGPA, linear-only MGPA and IGBP. Thus the maximum also reflects retained temporal structure. CORAL reaches $S = 1$ in both orientations under the common-offset interface; these are not results for token-specific CORAL maps.

Comparison with IGBP and task utility. IGBP and Iterative MGPA both attain full-bank $S = .8718$, with pooled-mean scores .7832 and .7954, respectively. IGBP has lower movement (.0956 versus .1490), slightly higher frozen-task AUROC (.6891 versus .6880), and similar refitted-task AUROC (.6884 versus .6886). Against identity, Iterative MGPA’s frozen/refitted task changes are -0.0002 [-.0017, $+0.0013$] and $+0.0004$ [-.0028, $+0.0034$].

C.4 P300-SELECTED REUSE: ABSOLUTE OUTCOMES AND PAIRED COMPARISONS

Tables 11–12 supplement the main reuse comparison at its unchanged P300-selected operating points. Both MGPA corrections improve transferred worst AUROC over LEACE on N170 and MMN, with paired intervals excluding zero; on P300 these intervals contain zero. Control-head changes from identity are small: Closed-form MGPA gains .0012 [.0006, .0018] on MMN, while the other three MGPA transfer-task intervals contain zero. N170 worst AUROC remains below .5: the gain is improved robustness rather than complete recovery.

Table 11: **Absolute reuse outcomes.** A/I/R: aligned/independent/reversed AUROC of the shortcut-sensitive head; worst/control results are in Table 4. Iterative entries average fits; other maps are evaluated once. CORAL/FEATMAP require device identity.

Method	P300			N170			MMN		
	A	I	R	A	I	R	A	I	R
Identity	0.770	0.568	0.355	0.914	0.547	0.122	0.753	0.603	0.446
LEACE	0.604	0.602	0.601	0.782	0.590	0.384	0.614	0.633	0.652
MGPA-CF	0.643	0.621	0.599	0.735	0.579	0.420	0.642	0.644	0.646
MGPA-Iter	0.636	0.617	0.599	0.744	0.594	0.441	0.632	0.639	0.646
CORAL	0.662	0.607	0.550	0.818	0.564	0.277	0.639	0.632	0.624
FEATMAP	0.625	0.600	0.573	0.697	0.591	0.471	0.623	0.621	0.618

Table 12: **Paired reuse gains.** Method-minus-comparator AUROC and 95% intervals, recomputing worst AUROC inside each bootstrap draw (Appendix B.4).

Task	Method	Reference	Δ worst [95% CI]	Δ control head [95% CI]
P300	MGPA-CF	Identity	+0.244 [+0.220, +0.268]	+0.0005 [−0.0010, +0.0022]
P300	MGPA-CF	LEACE	−0.002 [−0.022, +0.027]	+0.0050 [+0.0036, +0.0066]
P300	MGPA-Iter	Identity	+0.244 [+0.224, +0.262]	+0.0003 [−0.0016, +0.0025]
P300	MGPA-Iter	LEACE	−0.002 [−0.016, +0.021]	+0.0047 [+0.0034, +0.0062]
N170	MGPA-CF	Identity	+0.298 [+0.244, +0.340]	−0.0014 [−0.0044, +0.0019]
N170	MGPA-CF	LEACE	+0.036 [+0.005, +0.067]	+0.0002 [−0.0023, +0.0037]
N170	MGPA-Iter	Identity	+0.319 [+0.251, +0.380]	−0.0010 [−0.0035, +0.0013]
N170	MGPA-Iter	LEACE	+0.057 [+0.034, +0.083]	+0.0006 [−0.0009, +0.0025]
MMN	MGPA-CF	Identity	+0.195 [+0.168, +0.215]	+0.0012 [+0.0006, +0.0018]
MMN	MGPA-CF	LEACE	+0.028 [+0.002, +0.034]	+0.0024 [+0.0006, +0.0043]
MMN	MGPA-Iter	Identity	+0.186 [+0.157, +0.209]	+0.0003 [−0.0014, +0.0017]
MMN	MGPA-Iter	LEACE	+0.019 [+0.004, +0.024]	+0.0016 [−0.0005, +0.0036]

Across the three Iterative MGPA fits, worst-association AUROC ranges from .417 to .471 on N170 and .622 to .639 on MMN. Every P300-selected map, reused unchanged, improves this score over LEACE on both transfer tasks; Table 12 reports intervals for the mean across fits.

D REPRODUCING THE EXPERIMENTS

The [repository README](#) indexes code, fixed configurations and participant-level results, and provides two reproduction workflows. Reported statistics and figures can be rebuilt from supplied results without refitting; fresh experiments use separately acquired recordings and encoder weights. The guide links data-acquisition instructions, per-study protocols, and records of numerical provenance and validation scope.